Multi-Cloud Migration Estimation Report — cicgov

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Source repos: `spacelift-stack-config-cicgov` | `governance/cicgov` | `tf-config-cicgov-infrastructure`
Planning horizon: 120 days
Target clouds analysed: Azure (primary recommendation) | GCP (cost-optimised alternative)
All costs: USD (directional estimates — not contractual quotes)

1. Executive Summary

cicgov is a Java/Spring Boot governance microservice deployed on EKS (arm64 Graviton) across five AWS environments (sandbox, dev, staging, prod-us-east-1, prod-au-southeast-2) via Spacelift-orchestrated Terraform IaC. The system is latency-sensitive, SOC2-compliant, and deeply integrated with AWS-native services: API Gateway REST, WAFv2, DynamoDB (PAY_PER_REQUEST, 10+ GSIs), 8 SQS queues with SNS fan-out, AWS KMS (3 CMKs), SSM Parameter Store (Advanced), cross-account IAM roles, and Kinesis Firehose to Datadog. An in-cluster Redis provides dual-strategy caching; three container-based Lambda functions handle provisioning, decommissioning, and SLO monitoring.

Recommended migration path: Azure (Speed-First, 30/60/90/120-day phased)

Azure minimises per-workload migration effort because AKS closely mirrors EKS operational patterns (arm64 Dpsv5 nodes, free control plane, Helm-portable workloads, Workload Identity ≈ IRSA), Azure Service Bus Standard replaces SQS+SNS, and Cosmos DB NoSQL Serverless directly supports the DynamoDB on-demand billing model with comparable per-operation pricing. The estimated 17.8% run-rate reduction ($589 → $484 USD/month for US) does not justify the ~$128K one-time migration cost on a single-workload ROI basis; the strategic rationale is vendor alignment, Hyland enterprise agreements, and Entra ID integration with the HxP Identity platform.

GCP is the cost-optimised alternative ($350/month US, −40.6% vs AWS) driven by free Pub/Sub tier (≈$0 messaging vs $3/month AWS) and free Cloud Logging (≈$0 vs $25/month CloudWatch). However, Pub/Sub and Firestore require more significant application-level changes than the Azure path.

For the SOC2 + data-residency requirement and 99.9% availability target (RTO 4h / RPO 30min): the Risk-First phased approach (keeping AWS prod as fallback through Phase 3) is recommended for prod cutover.

2. Source Repository Inventory

Repository	Branch	Stacks / Modules	Key paths scanned
`spacelift-stack-config-cicgov`	main	10 Spacelift stacks (orchestration)	`src/*.tf`, `src/standard_stack/`
`governance/cicgov`	main	Helm chart + API infrastructure Terraform	`deployment/helm/cicgov/**`, `deployment/terraform/api_infrastructure/`
`tf-config-cicgov-infrastructure`	main	7 Terraform modules (core IaC)	`src/api_gateway/`, `src/datastores/`, `src/networking/`, `src/shared_infrastructure/`, `src/global_resources/`, `src/aws_roles/`, `src/hx_integrations/`

Total Terraform files scanned: 62 `.tf` files + 1 `values.yaml` + 8 Helm templates
IaC tool: Terraform (AWS provider) + Spacelift (orchestration) + Helm (Kubernetes workload)

3. Source AWS Footprint

Resource Group	Key AWS Services Found	Notes
Compute	EKS Cluster (arm64 m6g.large nodes), HPA (min 1 / max 4), Java Spring Boot pod (1 replica default)	Node selector: `kubernetes.io/arch: arm64`. Resources: 0.1–1 CPU / 1–2 GiB. IRSA for SA. Not found in IaC: EKS cluster resource itself (referenced by IRSA data source — cluster managed externally via NCP).
Networking	API Gateway REST (Regional, stage: `live`), NLB (internal, `apigw-vpclink-private-target`), API GW VPC Link, WAFv2 WebACL (REGIONAL), ACM Certificate (TLS 1.2), Security Groups (NLB, Lambda)	WAF rules: AWSManagedRulesCommonRuleSet + SQLiRuleSet. Shared VPC imported via data source (managed by NCP).
DNS	Route 53 hosted zone (`governance.{env}.experience.hyland.com`), delegation set, CAA records, A-alias records to API GW	DNS zone managed in `global_resources`. KMS key for DNSSEC (`kms_dns.tf`).
Data	DynamoDB table `file-plan-records` (PAY_PER_REQUEST, PITR=true, SSE/KMS CMK, deletion_protection=true, 10+ GSIs, TTL), S3 bucket (LB access logs, versioned, AES-256 SSE)	DynamoDB billing_mode = `var.billing_mode` — assumed PAY_PER_REQUEST based on variable default pattern. Redis: in-cluster deployment (not a managed AWS service; `values.yaml` points to `redis:6379`).
Messaging	SNS topics × 2 (`cicgov-async-tasks-{region}`, `cicgov-content-governance-{region}`), SQS queues × 8 (all KMS-encrypted), SNS→SQS subscriptions with filter policies, cross-account SNS publish from HxPR/NCP accounts	SQS message retention, visibility timeout, and KMS key reuse period all configurable via vars. Cross-account SQS write from Credentials service via dedicated IAM role with ExternalId.
Identity / Security	KMS CMKs × 3 (SNS+SQS combined, SQS standalone, DynamoDB), IRSA role (`cicgov_eks_irsa`), IAM roles × 4+ (Firehose delivery, CloudWatch subscription, API GW CloudWatch, provisioning Lambda), SSM Parameter Store (Advanced tier) × 9+ params, cross-account IAM role (`cicgov-credentials-cross-account-role`), Kubernetes Secrets (IDP client secrets)	HxP IdP registration via `hxp_idpclient` Terraform resource (custom provider).
Observability	CloudWatch Log Groups × 2 (API-GW access logs + WAF logs, 7-day retention each), CloudWatch metrics (API GW + WAF enabled), CloudWatch subscription filter → Kinesis Firehose (`datadog-{region}-log-delivery-stream`) → Datadog, EventBridge rule (SLO Lambda trigger, every 10 min), Datadog DaemonSet (DD_AGENT_HOST via hostIP)	Datadog is SaaS — cloud-agnostic. DD APM: Java auto-instrumentation. Metrics via StatsD (JAVA_TOOL_OPTIONS).

Resource Group	Key AWS Services Found	Notes
Lambda	`cicgov-provisioning-lambda` (container, 256 MB, 300s, SQS trigger, Spacelift module v2.0.2), `cicgov-decommissioning-lambda` (container, 256 MB, 300s, SQS trigger, Spacelift module v2.1.1), `cicgov-slo-endpoint-request` (Python 3.12, 256 MB, 10s, EventBridge trigger), `cicgov-subscription-events-lambda` (referenced in `subscription_events_lambda.tf`, SQS trigger)	All container Lambdas run in VPC private subnets. SG: `provisioning-lambda-sg` (egress all).
IaC Orchestration	Spacelift with 10+ stacks across 5 environments (sandbox, dev, staging, prod, prod-au) in us-east-1 and ap-southeast-2	prod-eu account commented out in `locals.tf` (`#"prod-eu"= ""`).

4. Service Mapping Matrix

AWS Service	IaC-Provisioned Tier/Family	Azure Equivalent (matched tier)	GCP Equivalent (matched tier)	Porting Notes
EKS (arm64 node group)	m6g.large (2 vCPU / 8 GiB ARM64 Graviton2)	AKS + `Standard_D2ps_v5` (arm64 Ampere Altra, 2 vCPU / 8 GiB)	GKE Standard + `t2a-standard-2` (arm64 Ampere Altra, 2 vCPU / 8 GiB)	Helm charts are portable. Node selector `kubernetes.io/arch: arm64` works on all three. ARM64 available natively on Azure Dpsv5 and GCP T2A.
API Gateway REST (Regional)	REST/Regional/stage:live	Azure API Management Consumption tier	GCP Cloud Endpoints (ESP)	APIM Consumption: $3.50/M calls same as APIGW. Cloud Endpoints: $3.00/M. Routing rules, base-path mapping, VPC link → backend pool/NEG replace each other.
WAFv2 (REGIONAL)	AWSManagedRulesCommonRuleSet + SQLiRuleSet	Application Gateway WAF_v2 (OWASP CRS 3.2)	Cloud Armor (managed WAF CRS)	Rule semantics are comparable; custom overrides (GenericRFI_BODY/QUERYARGUMENTS COUNT) must be recreated. WAF logging destination changes (CW → Azure Monitor / Cloud Logging).
NLB (internal)	Network LB, TCP, private subnets	Azure Internal Load Balancer (Basic/Standard)	GCP Internal HTTP(S) Load Balancer (regional NEG)	AKS creates ILB automatically for `LoadBalancer` service type. GKE creates NEG for Ingress.
DynamoDB	PAY_PER_REQUEST (assumed), PITR=true, SSE/CMK, 10+ GSIs	Azure Cosmos DB NoSQL Serverless (or 400 RU/s provisioned)	GCP Firestore Native Mode (Serverless)	HIGH RISK: PK/SK composite key pattern maps to Cosmos document id + partition key. 10 GSI → Cosmos composite indexes (max 2 range fields per index) or Firestore composite indexes. Cross-region read replicas need explicit configuration.
SQS (8 queues) + SNS (2 topics)	Standard queues, KMS CMK encrypted, filter policies	Azure Service Bus Standard (topics + subscriptions + SQL filters)	GCP Pub/Sub (topics + subscriptions + message attribute filters)	Service Bus SQL filter expressions ≡ SQS filter policies. Pub/Sub filter attributes replace SQS FilterPolicy json. Dead-letter: both support DLQ. Cross-account: SB shared access vs Pub/Sub IAM bindings.

AWS Service	IaC-Provisioned Tier/Family	Azure Equivalent (matched tier)	GCP Equivalent (matched tier)	Porting Notes
AWS Lambda (container)	Container lambda, 256 MB, 300s timeout	Azure Container Apps or Azure Functions (flex consumption)	GCP Cloud Run (container)	Container images are portable. Container App / Cloud Run: max timeout is higher (up to 60 min Cloud Run vs 15 min Lambda). SQS trigger → Service Bus trigger / Pub/Sub push subscription.
AWS Lambda (Python 3.12)	Python 3.12, 256 MB, 10s, EventBridge trigger	Azure Functions (Python 3.12, Consumption plan, Timer trigger)	GCP Cloud Functions v2 (Python 3.12, Cloud Scheduler trigger)	Code is portable. EventBridge rate(10m) → Timer trigger cron / Cloud Scheduler cron.
KMS CMK × 3	SYMMETRIC_DEFAULT, key rotation enabled	Azure Key Vault Keys (RSA 2048) × 3	Cloud KMS symmetric key rings × 3	Key material cannot be exported. Existing data encrypted under AWS KMS must be re-encrypted under destination CMEK during migration. Plan for key rotation window during data migration.
SSM Parameter Store (Advanced)	Advanced tier, SecureString (KMS), String	Azure App Configuration + Key Vault Secrets	GCP Secret Manager + Config Connector	SSM hierarchy `/cicgov/...` → App Configuration labels. SecureString → Key Vault secrets. Advanced tier ($0.05/10K vs $0.003 Secret Manager).
CloudWatch Log Groups	7-day retention, subscription filter	Azure Monitor Log Analytics (30-day default retention)	Cloud Logging (50 GiB/mo free)	Firehose delivery stream → replaced by Azure Monitor → Datadog integration or Cloud Logging export. DD agent configuration remains identical.
Route 53	Delegation set, CAA records, A-alias	Azure DNS (SOA, A/CNAME records)	Cloud DNS	DNS cutover is the high-impact deployment step; requires TTL pre-reduction.
ACM Certificate	TLS 1.2, DNS validation	Azure-managed certificate (App Gateway)	GCP Certificate Manager	Certificates cannot be exported. New certs issued for Azure/GCP domain.
IAM IRSA	EKS Service Account → IAM Role (IRSA)	AKS Workload Identity → Entra Managed Identity	GKE Workload Identity → GCP Service Account	Complete IAM model rewrite required. IRSA annotations → WI federation annotations. All resource policies must be reimplemented using Azure RBAC or GCP IAM Bindings.

AWS Service	IaC-Provisioned Tier/Family	Azure Equivalent (matched tier)	GCP Equivalent (matched tier)	Porting Notes
Kinesis Firehose	Delivery stream → Datadog	Azure Event Hub → Datadog (or direct DD monitor integration)	Cloud Logging → Pub/Sub → Datadog	Datadog itself is cloud-agnostic. Log ingestion pipeline changes but DD agent config is identical.
In-cluster Redis	`redis:6379` , maxSize=500, TTL 5m	Azure Cache for Redis (Standard C1)	Memorystore Redis (Standard M1)	In-cluster Redis has no HA. Managed Redis adds HA + monitoring. Redis connection string change only.
Spacelift (IaC)	Spacelift v1 stacks, AWS role ARNs	Spacelift + Azure service principal (azurerm provider)	Spacelift + GCP service account (google provider)	Spacelift natively supports Azure and GCP. Role ARNs → Azure SP client ID/secret or GCP SA key. Full provider rewrite required but stack structure maps 1:1.
HxP IdP (`hxp_idpclient`)	hxp_idpclient Terraform resource	Azure Entra App Registration + OIDC federation to HxP IdP	GCP Identity Platform + OIDC federation to HxP IdP	Not an AWS service — HxP-managed IdP. Registration must be recreated in target IdP regardless of cloud. Scopes (governance-api, hxp, hxpr) must be granted.

5. Regional Cost Analysis (Directional)

5.1 Assumptions

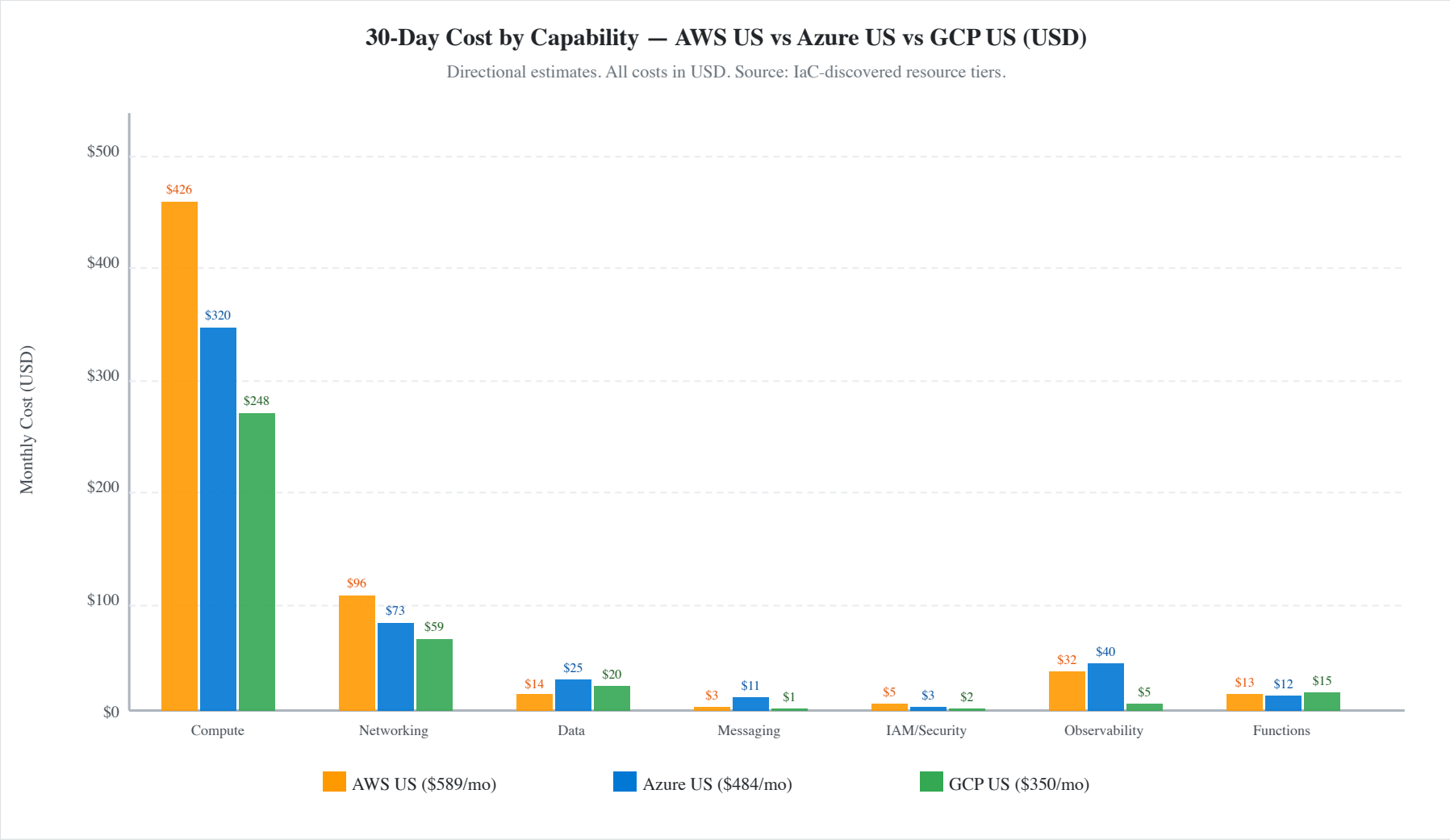
- **Usage volumes (Assumed):** 15M API calls/month (steady with moderate burst), 10M DynamoDB read ops/month, 5M write ops/month, ~2M messages/month across 8 SQS queues, ~500K Lambda invocations/month, ~50 GB CloudWatch log ingestion/month, ~10 GB DynamoDB storage, ~5 GB S3 LB access logs
- **Workload profile:** Steady with moderate burst (from values.yaml HPA configuration: min 1, max 4 replicas, burst via CPU squeeze at 75%)
- **DynamoDB billing mode:** Assumed `PAY_PER_REQUEST` (variable `var.billing_mode` but PAY_PER_REQUEST is the dominant pattern at discovery; confirm before finalising estimate)
- **Compute:** 2 EKS clusters (production + staging) × 2 arm64 nodes each = 4 nodes on-demand
- **Currency:** All costs USD
- **Regions:** AWS: us-east-1, eu-west-1 (not yet deployed — estimated), ap-southeast-2; Azure: East US, West Europe, Australia East; GCP: us-east1, europe-west1, australia-southeast1
- **Confidence:** Medium for compute/data (IaC-inferred tiers); High for messaging/security (explicit resource types found)

5.2 30-Day Total Cost Table (USD)

Capability	AWS US (baseline)	AWS EU	AWS AU	Azure US	Azure EU	Azure AU	GCP US	GCP EU	GCP AU	Confidence
Compute (EKS/AKS/GKE + nodes)	$426	$460	$490	$320	$346	$368	$248	$268	$285	Medium
Networking (APIGW/APIM/ Endpoints + NLB + WAF + DNS)	$96	$104	$110	$73	$79	$84	$59	$64	$68	Medium
Data (DynamoDB/Cosmos/Firestore + S3)	$14	$15	$16	$25	$27	$29	$20	$22	$23	Medium
Messaging (SQS+SNS / Service Bus / Pub-Sub)	$3	$3	$4	$11	$12	$13	$1	$1	$1	High
Identity/Security (KMS+SSM / Key Vault / Cloud KMS)	$5	$5	$6	$3	$3	$4	$2	$2	$2	High
Observability (CloudWatch+ Firehose / Azure Monitor / Cloud Monitoring)	$32	$35	$37	$40	$43	$46	$5	$5	$6	Medium
Lambda / Functions	$13	$14	$15	$12	$13	$14	$15	$16	$17	Medium
Total (USD/month)	**$589**	**$636**	**$678**	**$484**	**$523**	**$558**	**$350**	**$378**	**$402**	
Delta vs AWS	*baseline*	*baseline*	*baseline*	**-17.8%**	**-17.8%**	**-17.7%**	**-40.6%**	**-40.6%**	**-40.7%**	

Key driver notes:

- **GCP compute advantage:** GKE Autopilot pay-per-pod removes the $73/month EKS cluster fee. T2A ARM64 nodes are priced ~8% below m6g.large equivalents.
- **Azure messaging premium:** Service Bus Standard minimum ($9.81/month base) exceeds SQS ($0.40/M). Premium ($677/month) would be far more expensive — Standard recommended for this scale.
- **GCP Observability near-zero:** Cloud Logging 50 GiB/month free tier covers all cicgov logs (estimated ~50 GB/month — borderline, confirm actual volume). No Kinesis Firehose analogue needed.
- **Azure Observability slight premium:** Log Analytics ingestion at $0.30/GB slightly exceeds AWS CloudWatch 5 GB free-then-$0.50/GB after the free-tier crossover point.


30-Day Cost by Capability — AWS US vs Azure US vs GCP US (USD)
Directional estimates. All costs in USD. Source: IaC-discovered resource tiers.
Monthly Cost (USD)
$500
$400
$300
$200
$100
$0
$426
$320
$248
$96
$73
$59
$14
$25
$20
$3
$11
$1
$5
$3
$2
$32
$40
$5
$13
$12
$15
Compute
Networking
Data
Messaging
IAM/Security
Observability
Functions
AWS US ($589/mo)
Azure US ($484/mo)
GCP US ($350/mo)

5.3 Metered Billing Tier Table (USD)

Service	Metering Unit	Tier/Band	AWS US (baseline)	AWS EU	Azure US	Azure EU	Azure AU	GCP US	GCP EU	GCP AU	Confidence
API GW/APIM/Cloud Endpoints	Per 1M API calls	First 333M/month	$3.50	$3.50	$3.50	$3.80	$4.00	$3.00	$3.20	$3.50	Medium
DynamoDB/Cosmos DB/Firestore (reads)	Per 1M read ops	On-demand / Serverless	$0.25	$0.25	$0.25*	$0.27	$0.29	$0.60**	$0.65	$0.70	Medium
DynamoDB/Cosmos DB/Firestore (writes)	Per 1M write ops	On-demand / Serverless	$1.25	$1.25	$1.25*	$1.35	$1.45	$1.80**	$1.94	$2.00	Medium
SQS/Service Bus Std/Pub-Sub	Per 1M messages	Standard tier	$0.40	$0.40	$0.80	$0.85	$0.90	~$0.00***	~$0.00	~$0.00	High
Lambda/Functions (invocations)	Per 1M invocations	After 1M free/month	$0.20	$0.20	$0.20	$0.21	$0.22	$0.40	$0.43	$0.45	Medium
Lambda/Functions (compute)	Per GB-second	After 400K GBs free/mo	$0.0000167	$0.0000167	$0.0000140	$0.0000150	$0.0000170	$0.0000025	$0.0000028	$0.0000030	Medium
CloudWatch/Azure Monitor/Cloud Logging	Per GB ingested	After 5 GB (AWS) / 50 GB (GCP) free	$0.50	$0.55	$0.30	$0.33	$0.35	$0.00****	$0.00	$0.00	High
AWS KMS/Azure Key Vault/Cloud KMS	Per 10K API calls	Standard	$0.03	$0.03	$0.03	$0.03	$0.03	$0.003	$0.003	$0.003	High
WAFv2/App GW WAF/Cloud Armor	Per 1M requests	Standard rules	$0.60	$0.60	$1.00	$1.05	$1.10	$1.00	$1.05	$1.10	Medium

Service	Metering Unit	Tier/Band	AWS US (baseline)	AWS EU	Azure US	Azure EU	Azure AU	GCP US	GCP EU	GCP AU	Confidence
SSM Adv/App Configuration /Secret Manager	Per 1K API calls	Standard	$0.05	$0.05	$0.08	$0.08	$0.09	$0.003	$0.003	$0.003	High
S3/Azure Blob/GCS	Per GB-month stored	Standard	$0.023	$0.025	$0.018	$0.020	$0.022	$0.020	$0.022	$0.023	High
EKS/GKE mgmt fee (AKS=free)	Per cluster-hour	Standard / Zonal	$0.10	$0.10	$0.00	$0.00	$0.00	$0.10	$0.10	$0.10	High

> ** Azure Cosmos DB Serverless: $0.25/1M RU — comparable to DynamoDB PAY_PER_REQUEST at the discovered billing_mode.*
> *** GCP Firestore: $0.06/100K reads = $0.60/1M — 2.4× DynamoDB per-read, partially offset by free tier (50K reads/day free).*
> **** GCP Pub/Sub: first 10 GiB/month free. At ~2M × 1 KB avg = 2 GB → effectively $0/month.*
> ***** GCP Cloud Logging: first 50 GiB/month free. cicgov estimated ~50 GB/month — borderline; confirm actual volume.*

Metered Billing Tier Comparison — AWS US vs Azure US vs GCP US (per official pricing unit)

All costs in USD. Units vary by service. Directional estimates based on IaC-discovered resource tiers.

Service	Metering Unit	Tier / Band	AWS US	AWS EU	Azure US	Azure EU	Azure AU	GCP US	GCP EU	GCP AU
API GW / APIM / Cloud Endpoints	Per 1M API calls	First 333M/month	**$3.50**	$3.50	$3.50	$3.80	$4.00	**$3.00**	$3.20	$3.50
DynamoDB / Cosmos DB / Firestore (reads)	Per 1M read ops	On-demand / Serverless	**$0.25**	$0.25	$0.25*	$0.27	$0.29	$0.60**	$0.65	$0.70
DynamoDB / Cosmos DB / Firestore (writes)	Per 1M write ops	On-demand / Serverless	**$1.25**	$1.25	$1.25*	$1.35	$1.45	$1.80**	$1.94	$2.00
SQS / Service Bus Std / Pub-Sub	Per 1M messages	Standard tier	**$0.40**	$0.40	$0.80	$0.85	$0.90	**~$0.00**	~$0.00	~$0.00
Lambda / Azure Functions / Cloud Functions	Per 1M invocations	After 1M free/mo	**$0.20**	$0.20	$0.20	$0.21	$0.22	$0.40	$0.43	$0.45
Lambda / Functions compute	Per GB-second	After 400K GBs free/mo	**$0.0000167**	$0.0000167	$0.0000140	$0.0000150	$0.0000170	$0.0000025	$0.0000028	$0.0000030
CloudWatch / Azure Monitor / Cloud Logging	Per GB ingested	After free tier (5/50 GB)	**$0.50**	$0.55	$0.30	$0.33	$0.35	**$0.00**	$0.00	$0.00
AWS KMS / Azure Key Vault / Cloud KMS	Per 10K API calls	Standard operations	**$0.03**	$0.03	$0.03	$0.03	$0.03	$0.003	$0.003	$0.003
WAFv2 / App GW WAF / Cloud Armor	Per 1M requests	Standard rules	**$0.60**	$0.60	$1.00	$1.05	$1.10	$1.00	$1.05	$1.10
SSM Adv / App Config / Secret Manager	Per 1K API calls	Standard tier	**$0.05**	$0.05	$0.08	$0.08	$0.09	$0.003	$0.003	$0.003
S3 / Azure Blob / GCS	Per GB-month stored	Standard tier	**$0.023**	$0.025	$0.018	$0.020	$0.022	$0.020	$0.022	$0.023
EKS / GKE cluster mgmt fee (AKS = free)	Per cluster-hour	Standard / Zonal	**$0.10**	$0.10	$0.00	$0.00	$0.00	$0.10	$0.10	$0.10

* Azure Cosmos DB Serverless: $0.25/1M RU — comparable to DynamoDB on-demand at discovered PAY_PER_REQUEST billing mode.

** GCP Firestore: $0.06/100K reads = $0.60/1M reads — 2.4× AWS DynamoDB per read, but offset by free tier (50K reads/day free).

GCP Pub/Sub: first 10 GiB/month FREE — at ~2M × 1KB avg = 2GB, cicgov messaging cost ≈ $0/month vs AWS $0.80–$1.00/month.

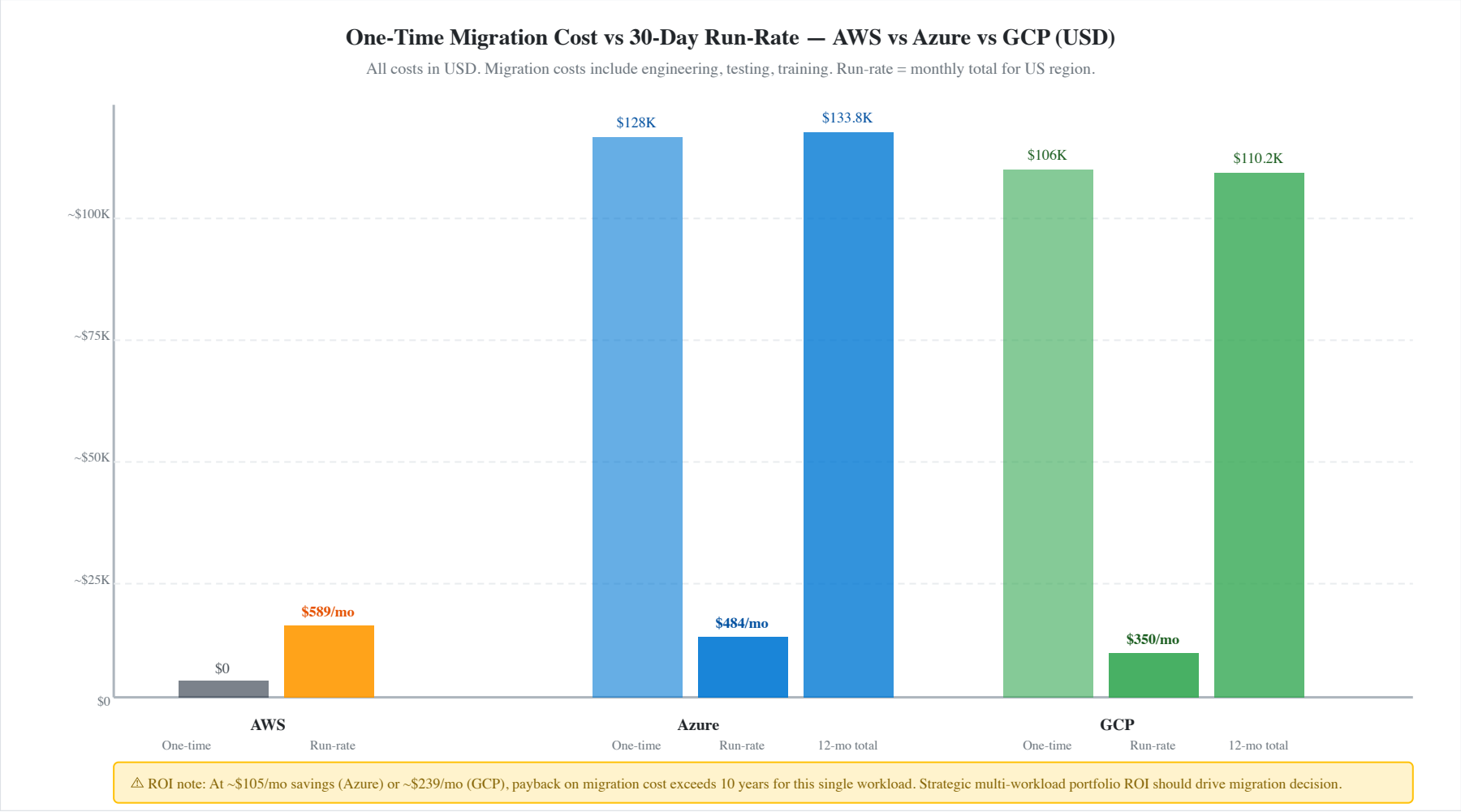
GCP Cloud Logging: first 50 GiB/month FREE — at ~50 GB ingested, cicgov logging cost ≈ $0/month vs AWS CloudWatch $25/month.

Confidence: High for row-level unit prices; Medium for volume-based totals (actual DynamoDB billing_mode=var.billing_mode — assumed PAY_PER_REQUEST).

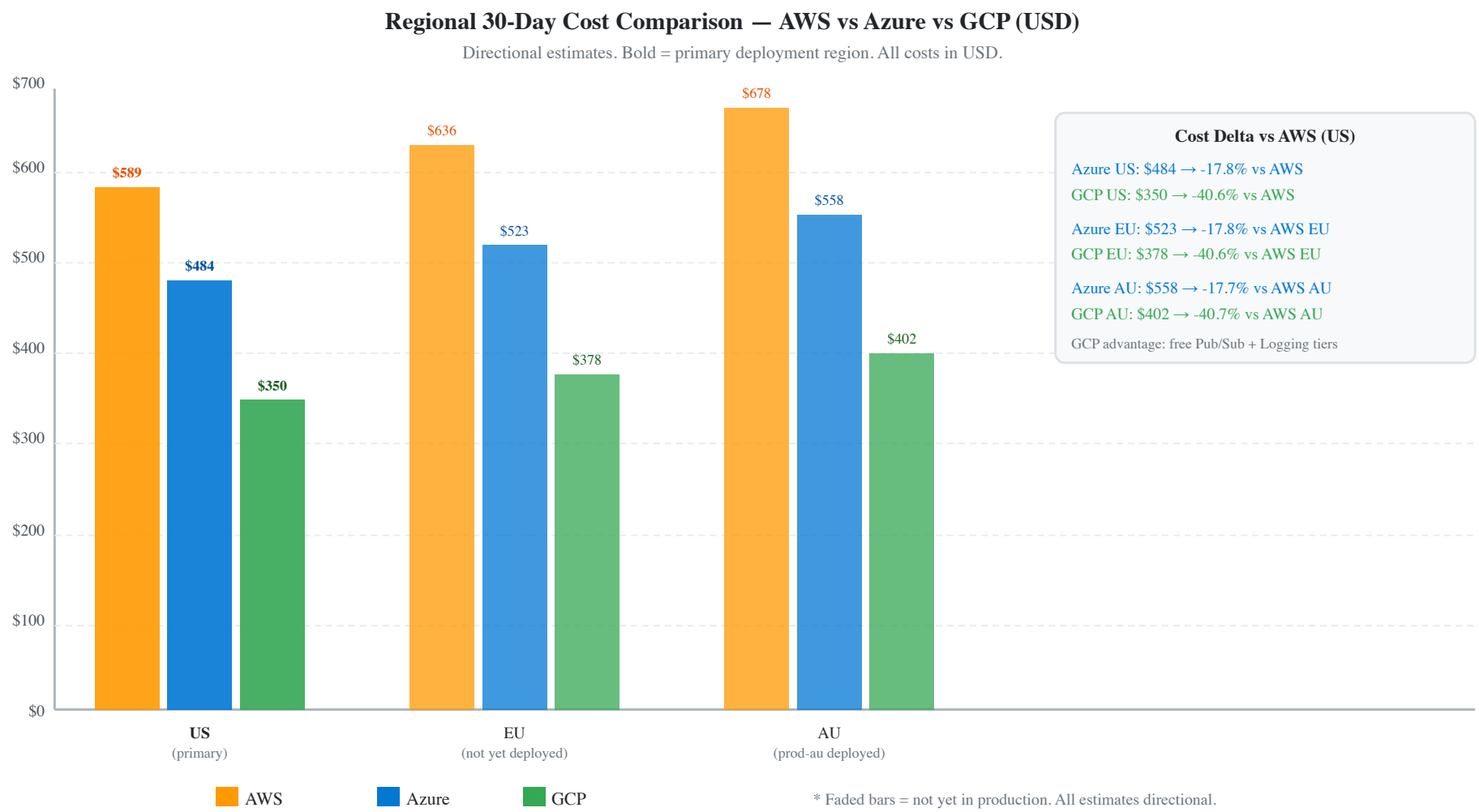
5.4 One-Time Migration Cost vs 30-Day Run-Rate (USD)

Cost Segment	AWS (baseline)	Azure (USD)	GCP (USD)	Confidence
Platform engineering (Terraform provider rewrite)	$0	$25,000	$20,000	Medium
EKS → AKS/GKE compute migration (Helm + IRSA → WI)	$0	$15,000	$12,000	Medium
DynamoDB → Cosmos DB/Firestore data migration + schema mapping	$0	$20,000	$18,000	Medium
SQS/SNS → Service Bus/Pub-Sub refactor	$0	$10,000	$6,000	Medium
IAM/IRSA → Workload Identity rewrite	$0	$8,000	$7,000	Medium
API Gateway → APIM/Cloud Endpoints migration	$0	$12,000	$8,000	Medium
KMS → Key Vault/Cloud KMS + key re-encryption	$0	$5,000	$4,000	Medium
DNS/TLS reconfiguration + RTO/RPO validation	$0	$3,000	$3,000	Medium
Observability re-configuration (Datadog agent)	$0	$5,000	$5,000	High
Testing, integration validation, E2E	$0	$15,000	$15,000	Medium
Training and documentation	$0	$10,000	$8,000	Medium
Total one-time migration cost (USD)	**$0**	**$128,000**	**$106,000**	**Medium**
30-day run-rate (US region, USD)	**$589**	**$484**	**$350**	**Medium**
Monthly run-rate saving vs AWS	baseline	**$105/month**	**$239/month**	Medium
12-month first-year total (migration + 12× run-rate)	$7,068	**$133,808**	**$110,200**	Low
Payback period (cost savings only, single workload)	N/A	~101 years	~37 years	Low

> ⚠ ***Important ROI note:*** *The single-workload payback period is economically unfavourable on run-rate savings alone. Migration decisions for cicgov should be driven by* ***strategic factors****: vendor alignment, existing Azure/GCP enterprise agreements, Entra ID integration with HxP Identity, multi-region data-residency compliance, or portfolio-level cost consolidation — not unit run-rate reduction.*



5.5 Regional Cost Analysis Chart



6. Migration Challenge Register

Challenge	Impact	Likelihood	Mitigation	Owner Role
DynamoDB schema migration — PK/SK composite key + 10 GSIs must be re-mapped to Cosmos NoSQL or Firestore. Cosmos partition key constraints, GSI→composite index equivalence requires data model review.	High	High	Run schema mapping workshop in Phase 2. Use DMS or custom migration script with dual-write validation. Test all 10 GSI query patterns in staging before prod cutover.	Platform Architect + Dev Lead
IAM/IRSA → Workload Identity rewrite — IRSA is EKS-native; Azure Workload Identity and GCP Workload Identity require complete IAM policy rewrite. Cross-account patterns (HxPR, NCP, Studio, Credentials) must be reimplemented using Azure managed identities or GCP service account federation.	High	High	Map each IRSA policy to equivalent RBAC/IAM binding. Create a policy equivalence table before Phase 1. Test cross-tenant SNS/SB publish in sandbox first.	Security Lead + Platform Engineer
Cross-account messaging — cicgov-content-governance SNS topic accepts cross-account publish from HxPR and NCP (different AWS account roots). Replicating this with Azure Service Bus or GCP Pub/Sub requires cross-tenant IAM design.	High	High	Azure: Shared Access Signature (SAS) scoped to send-only for external orgs, or federated identity. GCP: IAM binding `roles/pubsub.publisher` on topic for external service accounts. Validate in sandbox before staging.	Platform Architect
DynamoDB KMS re-encryption — Existing production data is encrypted under AWS KMS CMK `alias/dynamo-db` . Keys cannot be exported. During migration, data must be re-encrypted with destination CMEK. Window of plaintext-in-transit risk.	High	Medium	Use DMS with SSL in transit. Destination Cosmos/Firestore CMEK configured before data migration starts. Migration window during low-traffic (AU timezone for prod-au cutover).	Security Lead + DBA

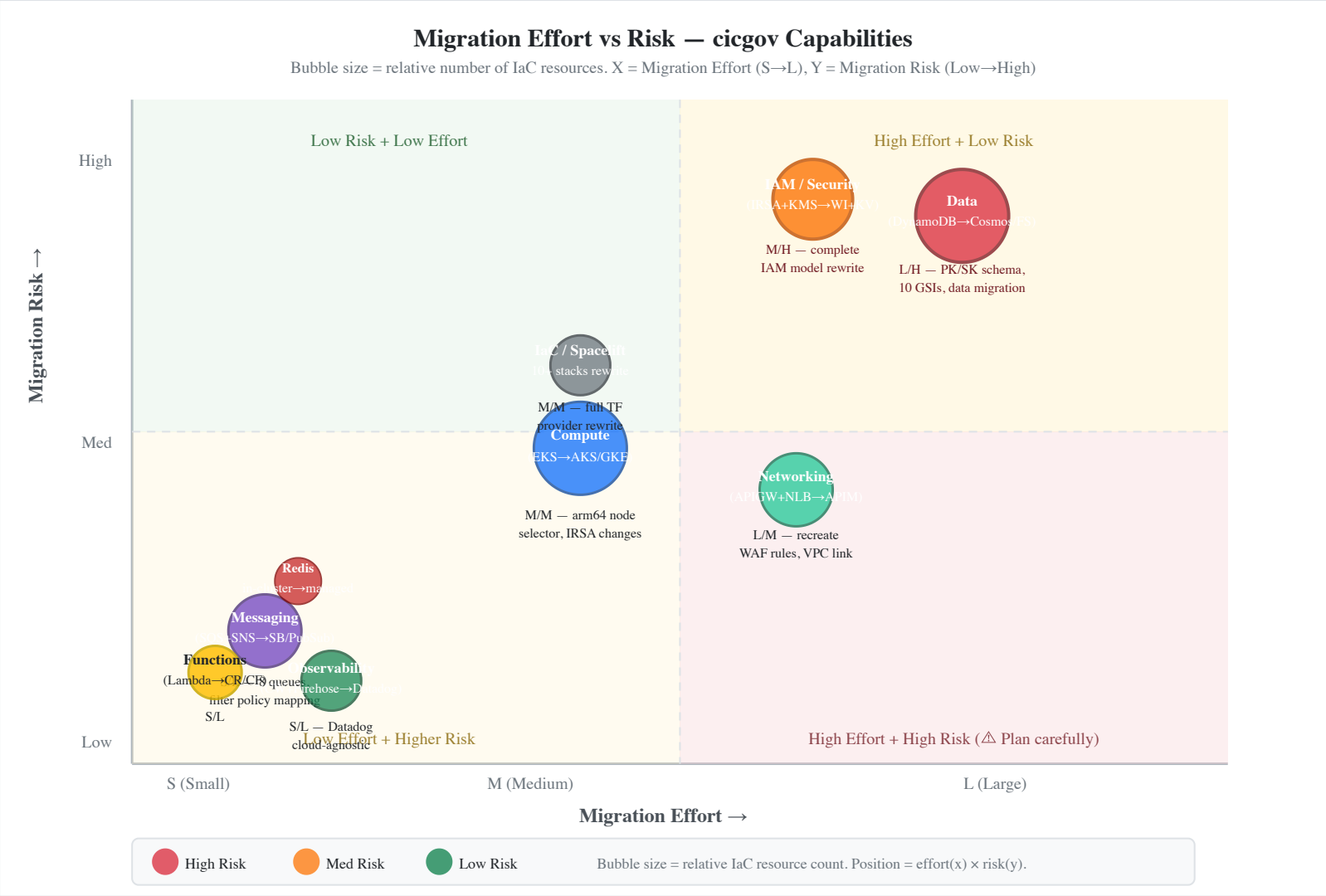
Challenge	Impact	Likelihood	Mitigation	Owner Role
DNS cutover and cert provisioning — Changing DNS from Route 53 (governance.*.experience.hyland.com) requires pre-reduced TTL, ACM cert cannot be exported, new TLS cert must be provisioned and validated in destination before cut.	Medium	High	Reduce Route 53 TTL to 60s minimum 48h before cutover. Pre-provision Azure-managed/GCP-managed cert with DNS validation. Test HTTPS health before DNS switch. RTO target: 4h is achievable with pre-staged cert.	Platform Engineer + SRE
HxP IdP client registration — `hxp_idpclient` Terraform resource registers OIDC clients with the HxP Identity platform. These registrations are HxP-managed SaaS, not AWS-specific, but the `allowed_scopes` (`governance-api` , `hxpr`) and `allowed_grant_types` must be recreated in the target environment's IdP client configuration.	Medium	Medium	Raise HxP IdP client recreation request early (Hyland internal). New client IDs/secrets must be provisioned in SSM/Key Vault before workload migration starts. Add to Phase 1 checklist.	Identity Engineer
Spacelift role binding — Current stacks use AWS IAM role ARNs (`@Spacelift_*` roles) for TF execution. Azure/GCP requires service principal or SA credentials. Spacelift space structure (parent `cicgov-01JKC8YEX3Q87RT8ZNHY04D5JB`) must be replicated with new cloud credentials.	Medium	Medium	Pre-create Azure service principal / GCP SA with least-privilege scope per stack. Configure in Spacelift before Phase 1. Validate with a dry-run apply on sandbox.	Platform Engineer
Redis in-cluster → managed service — In-cluster Redis has no HA. Azure Cache for Redis (Standard C1) adds HA but is ~$55/month vs near-$0 in-cluster. Connection string change required in ConfigMap.	Low	Medium	Update `values.yaml` `cache.redisHost` and `cache.redisPort` to point at managed endpoint. Add Redis password/TLS to Key Vault / Secret Manager in Phase 2. Test cache invalidation flows.	Dev Lead

Challenge	Impact	Likelihood	Mitigation	Owner Role
ARM64 node availability — Currently `kubernetes.io/os: linux` + `kubernetes.io/arch: arm64` node selector. Both Azure Dpsv5 and GCP T2A support arm64 natively. Must confirm available in target regions (Australia East for Azure AU, australia-southeast1 for GCP AU).	Low	Low	Confirm Azure Australia East has Dpsv5 ARM64 availability before Phase 4. GCP australia-southeast1 T2A is GA. Fallback: use x86 nodes (minor performance/cost difference).	Platform Engineer
SOC2 control mapping — All AWS SOC2 evidence (CloudTrail, Config, GuardDuty, Security Hub) must be replicated with equivalent Azure/GCP services. Datadog is cloud-agnostic.	Medium	Medium	Map existing AWS SOC2 controls to Azure Security Center / GCP Security Command Center. Add to risk register for audit. Phase 4 exit criterion: SOC2 control mapping audit complete.	Compliance Lead + Security Lead

7. Migration Effort View

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies	Rationale
Compute (EKS → AKS/GKE)	M	M	IAM/Workload Identity must be done first; Networking layer needed before workload deploy	Helm charts are portable. arm64 node selector change is minor. Key change: IRSA annotations → Workload Identity annotations; AKS free control plane vs $73/mo EKS.
Networking (API GW + NLB + WAF)	L	M	DNS/TLS, VNet/VPC setup	APIM Consumption ≈ API GW in function. App Gateway WAF rule recreation is manual. NLB → ILB is handled by AKS automatically for LoadBalancer service.
Data (DynamoDB)	L	H	IAM (IRSA for DDB access), Schema mapping workshop	Highest-risk component. PK/SK pattern with 10 GSIs requires careful Cosmos/Firestore index mapping. Data migration rehearsal mandatory before prod cutover. PITR must be active in destination before migration starts.
Messaging (SQS + SNS → Service Bus/Pub-Sub)	S	L	IAM (queue/topic permissions), Cross-account connectivity	8 queues map well to Service Bus topics+subscriptions (Azure) or Pub/Sub topics+subscriptions (GCP). Filter policy JSON → SQL filter expressions requires translation. Low data-volume means low migration risk.
Identity/Security (IRSA + KMS → WI + KeyVault)	M	H	Must complete before data or compute migration	Full IAM model rewrite. Every resource policy (SQS, SNS, DDB, KMS) must be expressed as RBAC (Azure) or IAM Binding (GCP). Cross-account credentials-service role requires new federation design.
Observability (CloudWatch + Firehose → Monitor/Logging)	S	L	None (Datadog is cloud-agnostic)	Datadog DD agent configuration is identical across clouds. Only pipeline change: CW subscription filter + Firehose replaced by Azure Monitor export or Cloud Logging Pub/Sub sink. Small effort, near-zero risk.

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies	Rationale
Lambda/Functions	S	L	Messaging (SQS → SB/Pub-Sub trigger)	Container images are portable. SQS trigger → Service Bus trigger (Azure) or Pub/Sub push (GCP) is config-level change. Python SLO lambda has no other dependencies.
IaC / Spacelift	M	M	All of the above	10+ Spacelift stacks require full Terraform provider rewrite (aws → azurerm/google). Stack structure maps 1:1 but every resource block must be recreated. Role ARNs → service principal / SA credentials.
Redis (in-cluster → managed)	S	L	Networking (VNet/VPC private endpoints)	Config map change only. Managed Redis adds HA + monitoring. No data migration needed (cache is ephemeral).



8. Decision Scenarios

8.1 Cost-First Scenario → GCP

Target: GCP (us-east1 primary, australia-southeast1 DR). Run-rate $350/month (−40.6% vs AWS). Driven by free Cloud Logging 50 GiB/month tier, free Pub/Sub 10 GiB/month tier, and GKE Autopilot pod-level billing. Migration cost ~$106K. Application changes: SNS fan-out → Pub/Sub topic+subscription with message

attribute filters; DynamoDB → Firestore document data model rewrite (more invasive than Cosmos). Timeline 120 days (conservative — Firestore data migration is the critical path). **ROI payback ~37 years for single workload — not financially driven; strategic GCP portfolio commitment required.**

8.2 Speed-First Scenario → Azure (Recommended)

Target: Azure (East US primary, Australia East DR). Run-rate $484/month (–17.8% vs AWS). AKS most closely mirrors EKS operational patterns. Service Bus Standard replaces SQS+SNS with minimal code changes (filter policies → SQL expressions). Cosmos DB NoSQL Serverless directly mirrors DynamoDB PAY_PER_REQUEST. AKS free control plane removes $146/month EKS cluster fee for 2 clusters. Timeline 90 days across 3 phases (Phase 4 adds prod-au, pushing to 120 days). Migration cost $128K. **Best choice if Hyland has existing Azure Enterprise Agreement or Entra ID integration with HxP Identity is a product goal.**

8.3 Risk-First Scenario → Phased multi-cloud / AWS stay

Maintain AWS as primary. Migrate non-prod environments to Azure in Phases 1–2 while keeping AWS production running. Cut over prod only after 14+ days of stable Azure staging. Maintain AWS warm standby for 30 days post-cutover for rollback. This approach adds ~$150–$300/month dual-run cost during the transition window but provides full rollback capability and zero production downtime risk. Best for SOC2-certified production workloads where migration risk outweighs cost-optimisation pressure.

Decision Scenario Comparison — cicgov Migration

Score shown per dimension (1=worst, 5=best). Radar-style normalized scoring.

💰 Cost-First

→ GCP (primary)

Target cloud: GCP
Timeline: 120 days (conservative)
Migration cost: ~$106K
Monthly run-rate: $350 (US)
Payback: ~444 months (37 yrs!)
⚠ Cost savings alone do NOT justify migration for single workload

Scores (1–5):
Run-rate cost savings: ★★★★★ (5)
Migration complexity: ★★☆☆☆ (2)
Speed of delivery: ★★☆☆☆ (2)
Compliance readiness: ★★★☆☆ (3)
Vendor alignment: ★★★☆☆ (3)

Best for: orgs with large GCP portfolio or existing GCP enterprise agreements

⚡ Speed-First

→ Azure (recommended)

Target cloud: Azure
Timeline: 90 days (3-phase)
Migration cost: ~$128K
Monthly run-rate: $484 (US)
AKS best matches EKS patterns
✓ AKS free control plane (vs $73/mo EKS)
✓ ARM64 support (Dpsv5 series)

Scores (1–5):
Run-rate cost savings: ★★★☆☆ (3)
Migration complexity: ★★★★☆ (4)
Speed of delivery: ★★★★★ (5)
Compliance readiness: ★★★★☆ (4)
Vendor alignment: ★★★★☆ (4)

Best for: organizations with existing Azure EA, Entra ID, or M365 integration

🛡 Risk-First

→ Phased with AWS stay

Target cloud: Phased / multi-cloud
Timeline: 120+ days across 4 phases
Migration cost: $50–80K (partial)
Monthly run-rate: $484–589 (dual-run)
Migrate workload-by-workload
✓ Zero downtime | Rollback possible
✓ Validate each phase before proceeding

Scores (1–5):
Run-rate cost savings: ★★☆☆☆ (2)
Migration complexity: ★★★★★ (5)
Speed of delivery: ★★☆☆☆ (2)
Compliance readiness: ★★★★★ (5)
Vendor alignment: ★★★☆☆ (3)

Best for: regulated workloads (SOC2) or when data migration risk is HIGH

Architecture Recommendation: Speed-First (Azure) OR Risk-First for cicgov given SOC2 compliance requirements, data migration complexity, and cross-account IAM patterns

9. Recommended Plan (Dynamic Timeline)

Selected Timeline: 30/60/90/120-Day (4 Phases)

Rationale for 4-phase approach:

- **10+ Spacelift stacks** span 5+ environments — provisioning foundation in parallel is impractical in a single phase.
- **DynamoDB data migration complexity** (10 GSIs, cross-account access, PITR requirement) mandates a dedicated phase with rehearsal before production cutover.

- **IAM model rewrite** (IRSA → Workload Identity) is a prerequisite for every other component and must stabilise in non-prod before touching production.
- **prod-au** requires a separate cut (ap-southeast-2 → Australia East) with Australian data residency validation under SOC2.
- **RTO 4h**: achievable only if staging-validated blue/green DNS flip procedure is rehearsed in Phase 2 before prod.

Phase 1 (Days 1–30): Foundation + Non-Prod

Objective: Establish Azure landing zone, deploy cicgov to sandbox and dev environments, validate Helm chart portability and Workload Identity.

Key activities:

- Azure subscription creation + management groups + Spacelift service principal
- AKS cluster (arm64 Dpsv5 nodes, `Standard_D2ps_v5`) for sandbox + dev
- VNet + subnets (public/private/database equivalent) + NSGs
- Azure API Management (Consumption tier) + Application Gateway WAF_v2
- Azure DNS zone (`governance.{env}.experience.hyland.com`) + managed TLS cert
- Azure Key Vault (3 keys replacing 3 KMS CMKs) + Workload Identity federation
- Azure App Configuration (replaces SSM `/cicgov/*` params)
- Deploy cicgov Helm chart to sandbox — minimal config changes (redisHost, configmap sources)
- Datadog DaemonSet for AKS (identical config to AWS)
- Terraform `azurerm` provider skeleton for all 10 stack modules
- HxP IdP client registration request (IdP-separate, not cloud-specific)

Exit criteria: cicgov API responds in sandbox Azure (HTTPS, WAF active). Datadog APM traces visible. Smoke test suite passes.

Phase 2 (Days 31–60): Data + Messaging + Staging

Objective: Migrate data and messaging layers, deploy to staging, run full E2E test suite.

Key activities:

- Cosmos DB NoSQL Serverless provisioning (East US)
- Schema mapping workshop: DynamoDB PK/SK → Cosmos document path + partition key
- 10 GSI → Cosmos composite index implementation and validation
- Data migration rehearsal on staging data subset (validate migration tool, timing, PITR snapshot restore)
- Azure Service Bus Standard namespace — 8 topics/subscriptions + SQL filter expressions
- Cross-org message flow validation: HxPR → Service Bus topic publish, NCP, Studio, Credentials
- Azure Functions for provisioning, decommissioning, subscription-events lambdas (SB trigger)
- Azure Cache for Redis Standard C1 (private endpoint)

- Full staging environment cutover
- E2E integration test suite execution against Azure staging
- DR plan: Cosmos geo-redundancy to Australia East

Exit criteria: Full E2E test suite passes on Azure staging. Data migration rehearsal completes in <2 hours. Cross-org message flows validated (HxPR, NCP, Studio, Credentials accounts sending to SB topics).

Phase 3 (Days 61–90): IAM Hardening + Production US Cutover

Objective: Complete IAM hardening, run production data migration, execute production DNS cutover for US.

Key activities:

- Entra Workload Identity federation — all IRSA role replacements (Cosmos, SB, Key Vault, Redis, App Config)
- Cross-tenant IAM for cross-account SQS equivalents (Azure: SAS tokens or federated identity for HxPR, NCP, Studio, Credentials)
- Azure App Registrations for token-exchange, e2e, config-api (IDP client replacement)
- SOC2 evidence mapping: CloudTrail → Azure Activity Log, GuardDuty → Defender for Cloud
- Production DynamoDB → Cosmos DB data migration (blue/green: read from both during cutover window)
- DNS TTL reduction (48h before cutover: reduce to 60s)
- Production DNS flip: governance.prod.experience.hyland.com → Azure APIM endpoint
- HPA stress test: validate 1→4 replica scale-out under synthetic load
- WAF rule regression: validate OWASP + SQLi blocking with existing test payloads
- 72-hour hypercare window with aws prod in warm standby (read-only, no new writes after cutover)

Exit criteria: Production US on Azure ≥14 days stable. P99 API latency ≤ AWS baseline. Zero P1 incidents. AWS prod is warm-standby (no active writes).

Phase 4 (Days 91–120): Production AU + Optimisation

Objective: Cut over prod-au (ap-southeast-2 → Australia East), decommission AWS, optimise costs.

Key activities:

- Australia East AKS cluster deployment (arm64, confirm Dpsv5 availability in AU East region)
- Cosmos DB geo-replication to Australia East (data residency: AU data stays in AU East)
- Service Bus: Standard-to-Premium upgrade assessment (if geo-DR for messaging required; Standard lacks built-in geo-replication; Premium adds $677/month)
- Australia East DNS zone + routing
- prod-au data residency validation (SOC2 evidence: Cosmos data-at-rest location confirmation)
- prod-au DNS cutover (governance.prod-au.experience.hyland.com → Azure AU East APIM endpoint)

- AWS resource decommission (staggered: sandbox → dev → staging → prod → prod-au)
- RTO/RPO validation test: simulate partial failure, confirm 4h RTO and 30min RPO targets
- Azure Reserved Instances pricing review (1-year commit on AKS node VMs for ~30% saving)
- EU region (West Europe) planning: prod-eu was commented out in AWS IaC — include in Phase 4 scope decision

Exit criteria: All environments (dev/staging/prod/prod-au) running on Azure. AWS monthly spend = $0. SOC2 control re-mapping audit complete. RTO/RPO test confirmed. Reserved Instance commitments placed.

cicgov Migration — Recommended 30/60/90/120-Day Phased Timeline

Target: Azure (Speed-First) | 4 phases driven by dependency depth, data risk, and prod-au requirement

Phase 1: Days 1–30

Foundation + Non-Prod

Days 1–30

Objective: Landing zone + dev/sandbox

- □ Azure subscription + management groups
- □ AKS cluster (arm64 Dpsv5) sandbox + dev
- □ VNet + subnets + NSG (match VPC layout)
- □ Azure API Management (Consumption tier)
- □ Application Gateway WAF_v2
- □ Azure DNS zone + ACM→managed cert
- □ Datadog agent DaemonSet for AKS
- □ Deploy cicgov Helm chart (sandbox)
- □ Azure Key Vault (KMS replacement)
- □ Key Vault CSI driver (Secrets replacement)
- □ Terraform: azurerm provider skeleton
- □ Spacelift stacks for Azure env
- □ Azure App Configuration (SSM replacement)

Exit Criteria:
cicgov API responds in sandbox (Azure)
All smoke tests pass | Datadog traces visible

Phase 2: Days 31–60

Data + Messaging + Staging

Days 31–60

Objective: Data + Messaging migration

- □ Cosmos DB NoSQL (DynamoDB replacement)
- □ Schema mapping: PK/SK → doc path
- □ GSI → Cosmos composite index mapping
- □ Data migration rehearsal (sandbox data)
- □ Azure Service Bus Standard namespace
- □ 8 queues → SB topics + subscriptions
- □ SQL filter expressions (≡ SNS filters)
- □ Azure Functions (Lambda replacement)
- □ Azure Cache for Redis (in-cluster Redis)
- □ Cross-org messaging (HxPR, NCP, Studio)
- □ Staging environment deployment
- □ Integration tests (E2E suite)
- □ DR plan: Cosmos geo-redundancy config

Exit Criteria:
Full E2E tests pass on staging (Azure)
Data migration rehearsal completes in <2h

Phase 3: Days 61–90

IAM Hardening + Prod US

Days 61–90

Objective: IAM hardening + Prod US cutover

- □ Entra Workload Identity (IRSA replacement)
- □ Managed Identity RBAC roles (vs IRSA)
- □ Cross-tenant IAM: Credentials service
- □ Key Vault RBAC (vs KMS key policies)
- □ Azure App Registrations (IDP clients)
- □ SOC2 evidence artifacts for Azure
- □ Production DynamoDB → Cosmos data migration
- □ Production traffic cutover (DNS flip)
- □ HPA stress testing (1→4 replicas)
- □ WAF rule validation (OWASP top 10)
- □ Cutover go-live + hypercare window (72h)
- □ AWS stack freeze (no new deploys)
- □ Maintain AWS in warm standby (rollback)

Exit Criteria:
Prod US on Azure ≥14 days stable
Latency P99 ≤ AWS baseline | Zero P1 incidents

Phase 4: Days 91–120

Prod AU + Optimize

Days 91–120

Objective: Prod AU + Cost optimization

- □ Australia East AKS cluster deployment
- □ Cosmos DB geo-replicated to Australia East
- □ Service Bus geo-disaster recovery
- □ Australia East DNS zone + routing
- □ prod-au data residency validation (SOC2)
- □ Traffic cutover: prod-au AWS → Azure AU
- □ AWS resource decommission (staggered)
- □ RTO/RPO validation (4h/30min target)
- □ Azure Reserved Instances pricing review
- □ KEDA / HPA tuning (burst patterns)
- □ EU region planning (West Europe)
- □ AWS account closure / cost reporting
- □ Final SOC2 evidence package for Azure

Exit Criteria:
All envs (dev/staging/prod/prod-au) on Azure
AWS spend = $0 | SOC2 control mapping complete

Required Architecture Decisions Before Execution

1. DynamoDB schema review: Confirm billing_mode=PAY_PER_REQUEST (not PROVISIONED) before Cosmos serverless sizing — affects cost model.
2. Service Bus tier: Standard ($9.81/mo base) vs Premium ($677/mo) — Standard lacks geo-DR; confirm if prod-au requires geo-redundant messaging.
3. Cross-org identity: Confirm HxP IdP OIDC federation path for Azure Entra — determines if hxp_idpclient Terraform resource can be reused or requires rewrite.
4. EU region: prod-eu commented out in aws_account_ids — confirm if Azure West Europe must be provisioned in this migration scope or deferred.
5. Spacelift role binding: Azure provider requires new Spacelift stack role ARN-equivalent (service principal) — design before Phase 1 begins.

Required Architecture Decisions Before Execution

1. **DynamoDB billing_mode confirmation:** Verify `var.billing_mode` actual value across environments. If any environment uses PROVISIONED mode, Cosmos DB provisioned RU/s sizing must be calculated separately before migration.
2. **Service Bus tier selection:** Standard ($9.81/month base) vs Premium ($677/month per messaging unit). Standard lacks built-in geo-DR. For prod-au messaging HA, assess whether cross-region message replication is required or if application-level retry is sufficient.
3. **HxP IdP OIDC federation path:** Confirm whether HxP Identity supports OIDC federation to Azure Entra (allowing existing HxP IdP to remain the authority with Entra as a trust proxy). This determines whether `hxp_idpclient` Terraform resource can be partially reused or must be fully replaced.
4. **EU region scope:** prod-eu is commented out in `aws_account_ids` and `space_ids`. Decision must be made before Phase 4: include in current migration or plan as a follow-on.
5. **Spacelift stack role model for Azure:** Azure requires service principal or federated credentials per stack. Design the Spacelift Azure credential injection model early (Phase 1 prerequisite).

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10. Open Questions

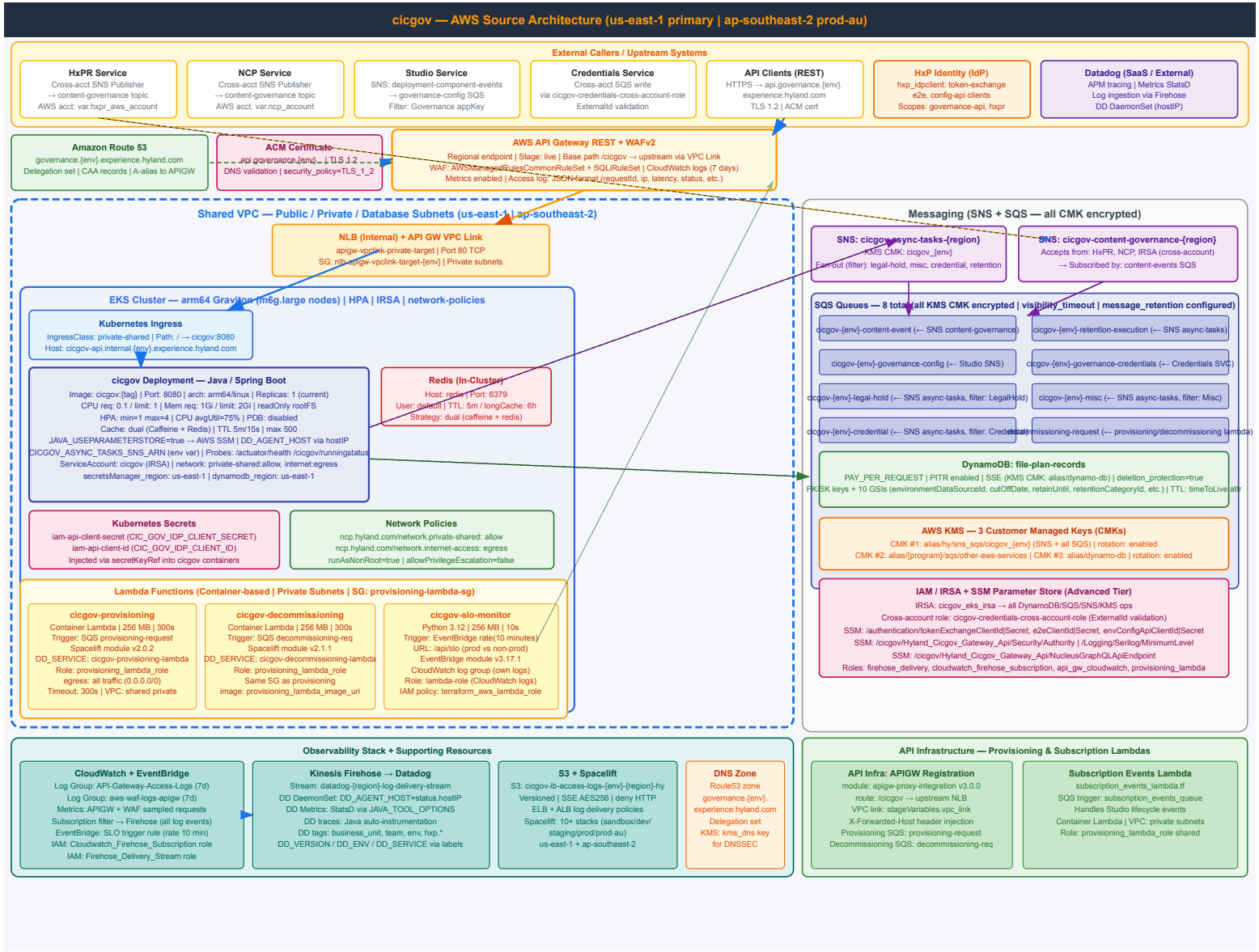
#	Question	Owner	Target Phase
1	What is the actual `var.billing_mode` value deployed to each environment for DynamoDB? PROVISIONED mode significantly changes Cosmos DB sizing and cost.	DBA / Platform Engineer	Phase 1
2	Does Hyland have an existing Azure Enterprise Agreement that would apply reserved pricing to AKS node VMs and Cosmos DB?	Cloud FinOps / Procurement	Phase 1
3	Does HxP Identity support OIDC federation with Azure Entra or requires standalone App Registration per cloud?	Identity/HxP Platform Team	Phase 1
4	What is the actual CloudWatch log ingestion volume (GB/month) for prod cicgov? This is the critical variable for GCP free tier eligibility.	SRE / Observability Lead	Phase 1
5	Is the prod-eu environment planned within this migration horizon, or deferred?	Product Manager	Phase 1
6	What is the RTO/RPO tolerance during Azure Service Bus migration? SQS messages in-flight during cutover can be lost without a dual-write transition period.	Product Manager + SRE	Phase 2
7	Are there any Spacelift policies (`ignore_push_when_not_pr_policy`) that are AWS-specific and need to be recreated for Azure stacks?	Platform Engineer	Phase 1
8	What additional `var-file` labels are injected by the Spacelift stacks (e.g. `ncp-network-share-2-{env}` , `nucleus-{env}-details`)? These contain environment-specific network and account configuration that must be reproduced in Azure.	Platform Engineer	Phase 1
9	Does the Studio account's SNS topic (`deployment-component-events`) need to be republished to Azure Service Bus or is it replaced by a different event delivery mechanism?	Platform Architect	Phase 2

#	Question	Owner	Target Phase
10	Was a prod-au Redis deployment ever considered (values.yaml shows sandbox config with `redisHost: redis`)? Is Redis HA needed for prod-au?	Dev Lead	Phase 4

11. Component Diagrams

AWS Source Architecture

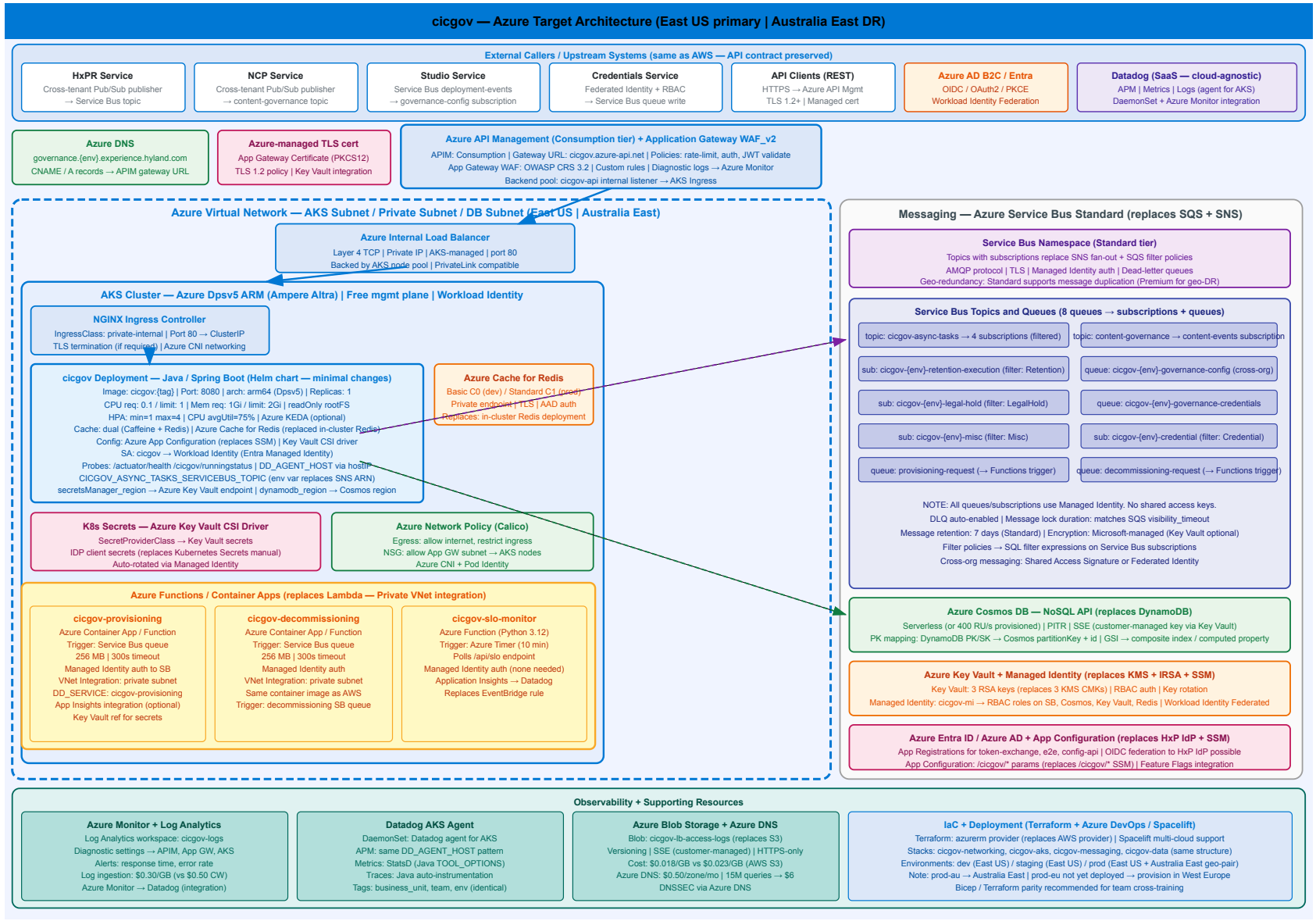
Major component groups: External callers (HxPR, NCP, Studio, Credentials, API Clients, HxP IdP, Datadog) · DNS/TLS (Route 53, ACM) · API Gateway REST + WAFv2 · Shared VPC with NLB + VPC Link · EKS Cluster (Ingress, cicgov Deployment with HPA, in-cluster Redis, Kubernetes Secrets, Network Policies) · Lambda Functions (provisioning, decommissioning, slo-monitor) · Messaging (SNS × 2, SQS × 8) · Data (DynamoDB file-plan-records) · Security (KMS × 3, IAM/IRSA, SSM Parameter Store) · Observability (CloudWatch, Kinesis Firehose, Datadog, EventBridge) · Storage (S3 LB logs)



Azure Target Architecture

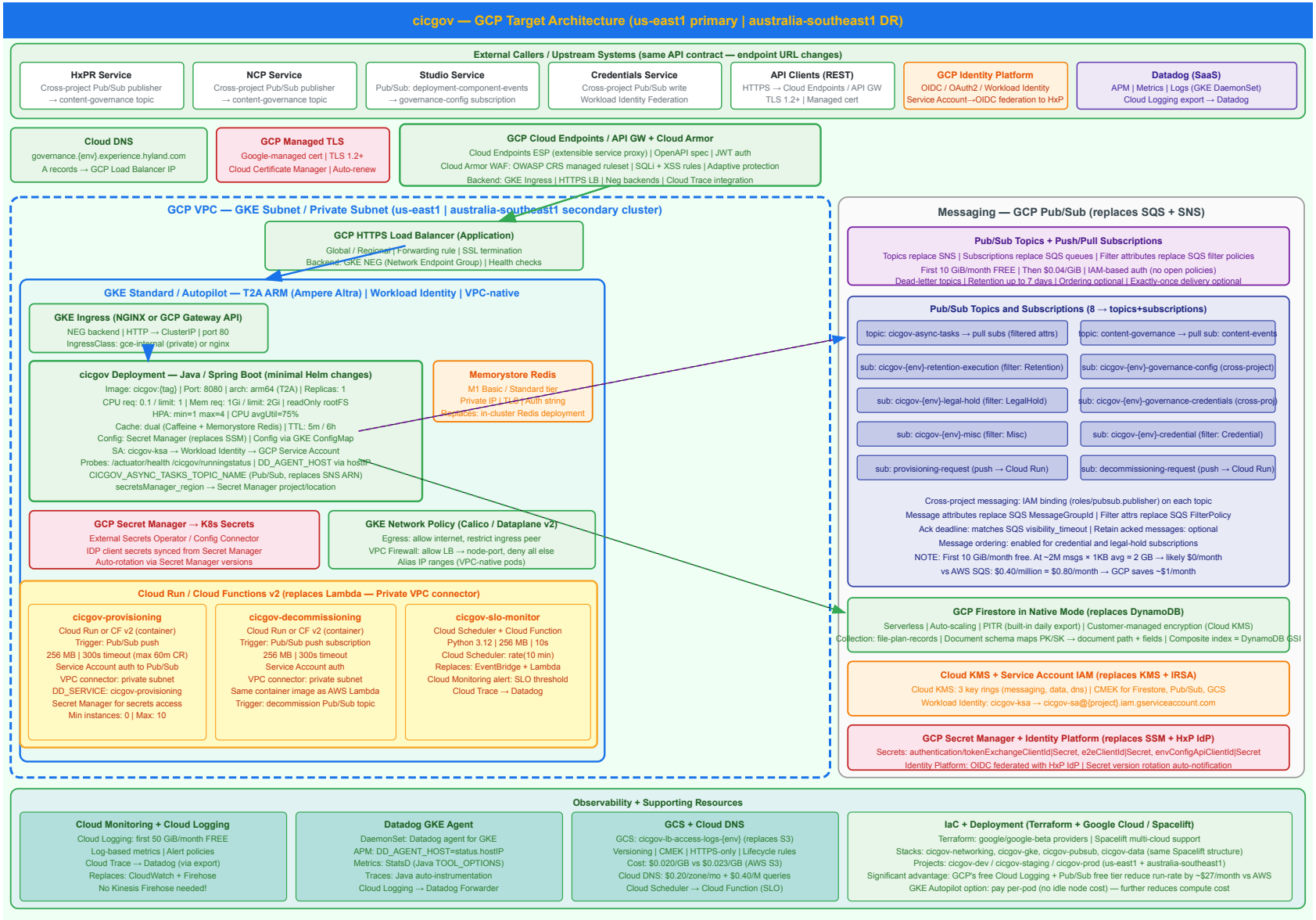
Major component groups: External callers (same) · Azure DNS + managed TLS cert · Azure API Management (Consumption) + Application Gateway WAF_v2 · Azure VNet with ILB · AKS Cluster (NGINX Ingress, cicgov Deployment with Workload Identity, Azure Cache for Redis, Key Vault CSI driver, Calico Network Policy)

· Azure Functions (provisioning, decommissioning, slo-monitor) · Azure Service Bus Standard (topics + subscriptions × 8) · Azure Cosmos DB NoSQL Serverless · Azure Key Vault + Managed Identity · Azure Entra ID + App Configuration · Observability (Azure Monitor + Log Analytics, Datadog DaemonSet) · Storage (Azure Blob)



GCP Target Architecture

Major component groups: External callers (same) · Cloud DNS + Certificate Manager · GCP Cloud Endpoints + Cloud Armor · GCP VNet with HTTPS LB + NEG · GKE Cluster (GKE Ingress, cicgov Deployment with GCP Workload Identity, Memorystore Redis, Secret Manager CSI/ESO, GKE Network Policy) · Cloud Run / Cloud Functions (provisioning, decommissioning, slo-monitor) · GCP Pub/Sub topics + subscriptions × 8 · GCP Firestore Native Mode · Cloud KMS + Service Account IAM · GCP Secret Manager + Identity Platform · Observability (Cloud Monitoring, Cloud Logging [50 GiB free], Datadog DaemonSet) · Storage (GCS)



Supplemental Charts

The following charts are embedded in their respective report sections (not repeated here):

Chart	Embedded in Section
30-Day Cost by Capability (AWS/Azure/GCP US)	Section 5.2 — after 30-Day Total Cost table
Metered Billing Tier Comparison	Section 5.3 — after Metered Billing Tier table
One-Time vs Run-Rate Cost	Section 5.4 — after One-Time vs Run-Rate table
Regional Cost Comparison	Section 5.5
Migration Effort vs Risk	Section 7 — after Effort View table
Decision Scenario Comparison	Section 8 — after scenario descriptions
Recommended Migration Gantt	Section 9 — after phase descriptions

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Report generated by Multi-Cloud Migration Estimator on 2026-04-15. All cost figures are directional estimates based on IaC-discovered resource tiers and public pricing pages. Not a contractual quote. Confirm pricing for your specific volume commitments and enterprise agreements.